

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination, 2023-24

Paper No. STB-101: Statistical Methods and Probability

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) What do you mean by the measure of central tendency? Define arithmetic mean and also write down its merits and demerits. [#1]
- (b) Derive the formula for the arithmetic mean of a composite series having two component series. Next, the average salary of male employees in a firm was Rs 5,200 and that of females in the same firm was Rs 4,200. Find the percentage of male and female employees. [#1]
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Question 2.

- (a) Define various partition values. Also, explain the procedure of how these can be located graphically. [#1]
- (b) Define standard deviation and root mean square deviation. Also, derive a relation between standard deviation and mean deviation of a series. [#1]
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Question 3.

- (a) Define r -th moment about origin and r -th moment about mean. Also, derive the relation between moments about mean in terms of moments about origin. [#1]
- (b) For a distribution, the mean is 10, variance is 16, $\gamma_1 = +1$, $\beta_2 = 4$. Obtain the first four moments about the origin. [#1]
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Question 4.

- (a) Give the mathematical definition of probability. Next, let an urn contain 6 white, 4 red, and 9 black balls. If 3 balls are drawn at random, find the probability that:
- (i) two of the drawn balls are white;
 - (ii) one is of each colour;
 - (iii) none is red;
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(iv) at least one is white.

[#1]

- (b) Define conditional probability and independence of events. Also, state and prove the multiplication theorem of probability for any two independent events. [#1]

Question 5.

- (a) A and B throw alternately with a pair of balanced dice. A wins if he throws a sum of six before B throws a sum of seven; while B wins if he throws a sum of seven before A throws a sum of six. If A begins the game, then find the probability of his winning the game. [#1]
- (b) If $A_1, A_2, \dots, A_n$ be any n mutually exclusive events with $P(A_i) > 0$ for all $i = 1, 2, \dots, n$, and let A be any event such that $A \subset \bigcup_{i=1}^n A_i$ with $P(A) > 0$, then find the value of $P(A_i|A)$ for all $i = 1, 2, \dots, n$. [#1]

Question 6.

- (a) Let the probability density function (p.d.f.) of a random variable X be:

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 2 \\ a(3-x), & 2 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$$

Determine the value of a and also compute $P(X < 1.5)$. [#1]

- (b) The joint p.d.f. of a two-dimensional random variable (X, Y) is given by:

$$f(x, y) = \begin{cases} 8xy, & 0 < x < 1, 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$

Obtain the conditional p.d.f.s of X given $Y = y$ and Y given $X = x$. Also check for the independence of X and Y . [#1]

Question 7.

- (a) Let the joint p.d.f. of X and Y be:

$$f(x, y) = \begin{cases} e^{-(x+y)}, & x > 0, y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Calculate $P(1 < X + Y < 2)$. [#1]

- (b) If $X_1, X_2, \dots, X_n$ be any n random variables and $a_1, a_2, \dots, a_n$ be any n constants, then find the value of $\text{Var}(\sum_{i=1}^n a_i X_i)$. [#1]

Question 8.

- (a) State and prove the addition theorem of expectation for any n jointly distributed random variables $Z_1, Z_2, \dots, Z_n$. [#1]

- (b) The joint probability distribution of two random variables X and Y is given by:

$$P(X = 0, Y = -1) = \frac{1}{2}, \quad P(X = 1, Y = -1) = \frac{1}{4}, \quad P(X = 1, Y = 1) = \frac{1}{4}$$

Then find:

- (i) the marginal probability distributions of X and Y ;
- (ii) the conditional probability distribution of X given $Y = 1$.

[#1]